Reproducing the manuscript results

Run every command below from the repository root. The scripts are deliberately named by their purpose so that each manuscript figure has one visible entry point.

Software

The analysis was prepared with R 4.4.2. Install the R dependencies and the local package once:

```
Rscript scripts/00_install_dependencies.R
```

The key non-base dependencies are `CVXR` with the `osqp` solver, `Matrix`, `foreach`, `doParallel`, `ggplot2`, `gridExtra`, `readr`, `glmnet`, `mgcv`, and `VariableScreening`. Solver installation can require system compilers on some platforms.

Recreate figures from archived results

The `results/` CSV files are the archived Monte Carlo outputs used by the plotting code (not rounded summaries). The following commands remake the corresponding figures without rerunning simulations:

```
Rscript scripts/figure1_main_simulation.R
Rscript scripts/figure2_variant_simulations.R
Rscript scripts/figure3_identifiable_setting.R
Rscript scripts/figure4_aerm_erm_bo.R
Rscript scripts/figureS1_mild_setting.R
```

Outputs are written under `figures/generated/`. The scripts also write machine-readable summary tables alongside each figure. The A-ERM/ERM/BO script uses the corrected A-ERM/ERM outputs in `results/main/` and the archived BO outputs in `results/bo/`. The plotting scripts retain their completeness checks and issue a warning when an archived CSV is incomplete; in that event they use the available replications, exactly as in the submitted figure-generation workflow.

Re-run the simulations

The full main and identifiable grids each contain 80 tasks; the additional simulations under varying IV strength, covariate dimension, and unmeasured confounding strength contain 60 tasks. The full settings use 200 Monte Carlo replications and 1,000 evaluation observations per task. They are intended for parallel/HPC execution. Each `run_*` script accepts the existing `--task-id N` command-line option. For example:

```
IVITR_N_REP=2 IVITR_N_EVAL=200 Rscript scripts/02_run_main_simulations.R --task-id 1
IVITR_N_REP=2 IVITR_N_EVAL=200 Rscript scripts/03_run_wald_main_simulations.R --task-id 1
IVITR_N_REP=2 IVITR_N_EVAL=200 Rscript scripts/04_run_variant_simulations.R --task-id 1
IVITR_N_REP=2 IVITR_N_EVAL=200 Rscript scripts/06_run_identifiable_simulations.R --task-id 1
```

Use scripts `05` and `08` for the corresponding Wald comparator runs, and scripts `07` and `08` for the mild setting. Newly generated files go to `generated/`; they do not overwrite archived manuscript results.

The outcome and treatment models for each simulation are documented in `SIMULATION_MODELS.md`. The source scripts retain the complete grid definitions and random seeds (`set.seed(i)`).

Public application demonstration

The application scripts expect variables `age`, `famhx`, `psa_bl`, `smoke`, `bmi`, `endPoint`, `trt`, and `adh`. The public synthetic dataset has exactly those fields and is generated deterministically:

```
Rscript scripts/01_prepare_synthetic_application_data.R
Rscript scripts/09_run_synthetic_application.R
Rscript scripts/figure5_synthetic_application.R
```

For laptop feasibility, script `09` defaults to a 300-subject fit, three folds, three bandwidths, and in-sample prediction. Set `IVITR_REAL_FIT_N`, `IVITR_REAL_NFOLD`, `IVITR_REAL_BW_LEN`, and `IVITR_REAL_PREDICT_ALL` before calling the script to change these settings for a full authorised-data run. For a one-threshold smoke test, also set `IVITR_REAL_D_SEQ=0.5`. If a numerical solver fails for a strict deferral setting, the public application runner records `fallback_zero_link` in `fit_status` and continues; this fallback is explicitly visible in the saved output.

To use an authorised copy of the original analytic data, set `PCPT_DATA_PATH` to a CSV with the same fields before running script `09`. Do not commit that file or the resulting participant-level outputs to a public repository.